

# Inputs

Date: 02/07/2026

Maker(s): Navya, Prince Patel  
Anon, Prince

Time: 1030hrs

Note Taker(s): Khushi Dhruvshat

## Chemicals used:

| Sr. No. | Chemical Name                  | % by Mass | Amount (in g) | Other Remarks (Particle Size, Hydration, Conditions kept in, Vendor) |
|---------|--------------------------------|-----------|---------------|----------------------------------------------------------------------|
| 1.      | KNO <sub>3</sub>               |           | 65.3          | ACS chemicals                                                        |
| 2.      | Sucrose                        |           | 26.98         | whole farm                                                           |
| 3.      | Al (325 mesh)                  |           | 5.05          | chemco, chemdyes corporation                                         |
| 4.      | Fe <sub>2</sub> O <sub>3</sub> |           | 2.9           | chemco, chemdyes corporation                                         |

## Casing Information

Description: Reusable metal casing with msea and metal nozzle & forward closure.

Material(s) Used:

Casing Dimensions (in mm):

OD: 3-43 cm ID: 2-58 cm IL: 12.77 cm OL:

Nozzle Diameter (in mm):

~~0.3 cm~~ 0.2 cm

Casing Mass (in g):

## Equipment Information

Utensil: non-stick pan

Mixing Instrument: spoon (wooden)

Others:

## Room Conditions:

Ambient Temperature:

Humidity:

Other Remarks: Fans off, window closed open

# Processing

| Time  | Chemical Added (Name, Amount) | Temperature Changes | Other Remarks |
|-------|-------------------------------|---------------------|---------------|
| 00:00 | All chemicals Added           | 130°C               |               |
| 2:30  | Clump Formation               | 130°C               |               |
| 3:45  | Temp down                     | 100°C               |               |
| 3:45  | Clay consistency              | 100°C               |               |
| 5:20  | Temp up                       | 130°C               |               |
| 7:10  | Temp down                     | 100°C               |               |
| 7:55  | Temp up                       | 130°C               |               |
| 13:45 | Pouring started               | 130°C               |               |
| 14:54 | Temp Down                     | 100°C               |               |
| 18:55 | Stove off                     | 0°C                 |               |

# Output

Final Motor Mass (in g): ~~445~~ 628.

Propellant Mass (in g): 76.

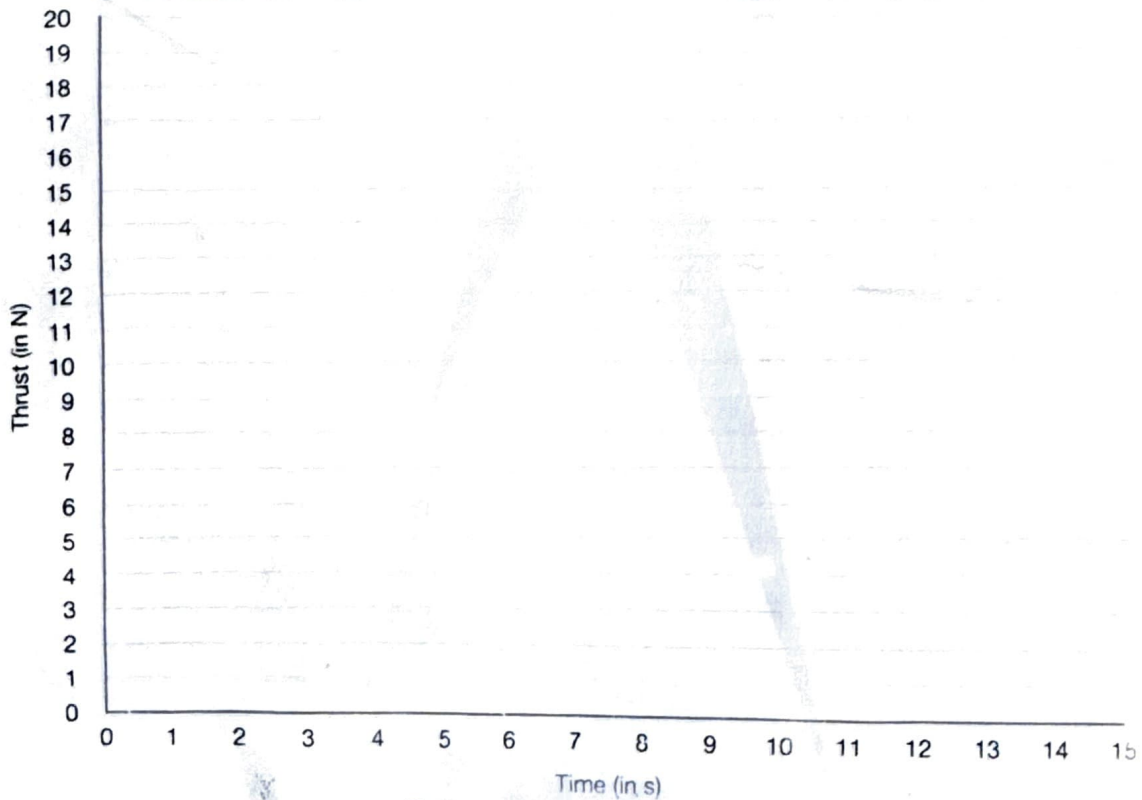
Burn Time (in s):

Burn Rate (in mm/s):

Impulse (in Ns):

Maximum Thrust (in N):

Thrust Curve:



Final Remarks: Tested on 03/07/2026

Namaste Sir,

Please allow us to test our motor in the 3rd period. I would like to assure you all safety protocols are followed with utmost priority.

②  
3/2/26